

TABLE OF CONTENTS

M-209 STATION PURIFIER PLANT

	Page No.
1. SCOPE OF WORK	M-209-2
2. CODES AND STANDARDS	M-209-2
3. DESIGN AND OPERATING CONDITIONS	M-209-3
3.1 Operating Requirements	M-209-3
3.2 Design Requirements	M-209-3
4. DETAILED REQUIREMENTS	M-209-4
4.1 Oil Purifier	M-209-4
4.2 Cleaning and Painting	M-209-5
4.3 Protection	M-209-5
4.4 Oil Storage Tanks	M-209-5
5. PERFORMANCE TESTING	M-209-5
5.1 Shop Test	M-209-5
5.2 Performance Test	M-209-5
6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)	M-209-6
7. INSTRUMENTATION AND CONTROLS	M-209-6
7.1 General	M-209-6
7.2 Local Control Panel	M-209-6
8. ELECTRICAL SPECIFICATIONS	M-209-7
8.1 General	M-209-7
8.2 Motors	M-209-7
FIGURE 1 STATION OIL PURIFIER P&I DIAGRAM	M-209-8

M-209 STATION OIL PURIFIER PLANT

1. SCOPE OF WORK

The Bidder shall supply an oil conditioning system for each Block in accordance with **Part 4, Subsection 1.2 “Summary of Work”** and as follow :

- One (1) clean oil storage tank, one dirty oil storage tank and one lube oil purifier.
- One (1) dirty lube oil transfer pump and one clean lube oil storage transfer pump.
- All integrated piping, piping supports, valves, fittings, heater, baseplate, bolts and necessary support steel.
- Complete instrumentation and control systems, devices, control panels, wiring, piping, tubing and valves.
- Complete electrical equipment for this system.
- Connection and facilities for future extension.

The Bidder shall state the price reduction in **Part 5 Schedule 2.1** in case the equipment specified in this section is deleted from the scope of work. The price reduction shall include equipment, erection, spare parts, etc.

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4 Section 2, Subsection 2.5**. The codes and standards of the following organizations or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this Specifications.

ANSI	American National Standards Institute.
API	American Petroleum Institute

ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standards
IEEE	Institute of Electronic and Electrical Engineers.
ISA	Instrument Society of America
JIS	Japan Industrial Standard
NEMA	National Electrical Manufacturer's Association
NFPA	National Fire Protection Administration Association

3. DESIGN AND OPERATING CONDITIONS

3.1 Operating Requirements

For the steam turbine generator unit the Station Oil Purifier Plant (oil transfer pump) shall be capable of drawing oil from the turbine oil reservoir and discharging to the clean or dirty oil storage tank. It shall be able to operate during shut down, stand by and under all operating conditions of the turbine generator.

3.2 Design Requirements

The system shall be capable of draining the turbine lube oil reservoir within three (3) hours and capable of filling the turbine lube oil reservoir oil tank within three (3) hours.

The system shall be able to clean and transfer the contents of the dirty oil storage tank to clean oil storage tank within eight (8) hours.

The system shall be of sufficient capacity to receive one (1) complete change of dirty oil plus a margin of 10 %, i.e the capacity of the dirty oil tank shall have a capacity equal to the steam turbine oil tank +10%.

The purifier equipment shall be designed to remove moisture, solid materials, and products of chemical decomposition of the oil.

4. DETAILED REQUIREMENTS

4.1 Oil Purifier

One (1) AC motor driven centrifugal oil purifier, for water and sludge removal from oil stored in the dirty oil tank shall be provided. The clean oil shall be stored in the clean oil tank.

The purifier shall be fully self-contained on an oil-tight baseplate with automatic lubrication as necessary. It shall be provided with all necessary pipe connections, including permanent connections for drainage to the dirty oil drains system and for supply of the sealing water to be taken from the feedwater make-up system.

For sludge disposal refer to Part 4 Section 3, M-404 Waste Water Treatment Plant.

Those parts of the purifier that are liable to corrosion, in particular the bowls, shall be manufactured from stainless steel.

Automatic shutdown and alarm facilities shall be provided to prevent the purifier from passing lubricating oil to waste.

The purifier shall be provided with an indirect electrical heating system with thermostatic control arranged to heat the incoming oil as necessary for optimum purifier performance.

4.2 Cleaning and Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and all external unmachined ferrous metal surfaces shall be painted in accordance with **Part 4, Section 3, S-420 and S-430**.

4.3 Protection

All exposed machined ferrous metal surfaces shall be coated with a suitable antirust compound before shipment.

All inlets and outlets shall be plugged or blanked off before shipment.

4.4 Oil Storage Tanks

Design and construction of the dirty and clean oil storage tanks shall be based on the recommendations of API Standard 650 (Welded Steel Tanks for oil storage), The National Fire Protection Association standards and acceptable industry practices and shall bear the Underwriters' Laboratory label.

Storage tank material shall be carbon steel for plate, pipe, castings, etc. The allowable stresses for ASME SA-285, Grade B or C, shall govern.

5. PERFORMANCE TESTING

5.1 Shop Test

All vapour extractors shall be shop tested.

All oil tanks shall be tested according to **Part 4, Section 3, M-012, "Shop Fabricated Tanks"**.

All pumps shall be shop performance tested and shop hydrostatically tested. Testing shall be in accordance with the Hydraulic Institute Standards.

5.2 Performance Test

The oil purifier shall be tested by the Bidder prior to shipment to verify that is performing according to the guarantee in every respect and shall conform to

the requirements of **Part 5, Schedules 7**. Field testing shall be performed to verify as-installed performance.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirements of **Parts 4, Subsection 2.7**.

7. INSTRUMENTATION AND CONTROLS

7.1 General

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control system and devices for the turbine lubricating oil purifier and pumps.

7.2 Local Control Panel

The Bidder shall furnish and install a local control lubricating oil purifier. The local control panel shall monitoring device which do not need to be installed location, and all operator interfaces such as indicators, control switches and alarms.

Refer to **Part 4, Section I-002, “Local Control Panel”** for the design requirement for the local control panels, **Part 4, Section 3, I-001, “Instrumentation and Control”** for the design requirements for Instrumentation and Control.

8. ELECTRICAL SPECIFICATIONS

8.1 General

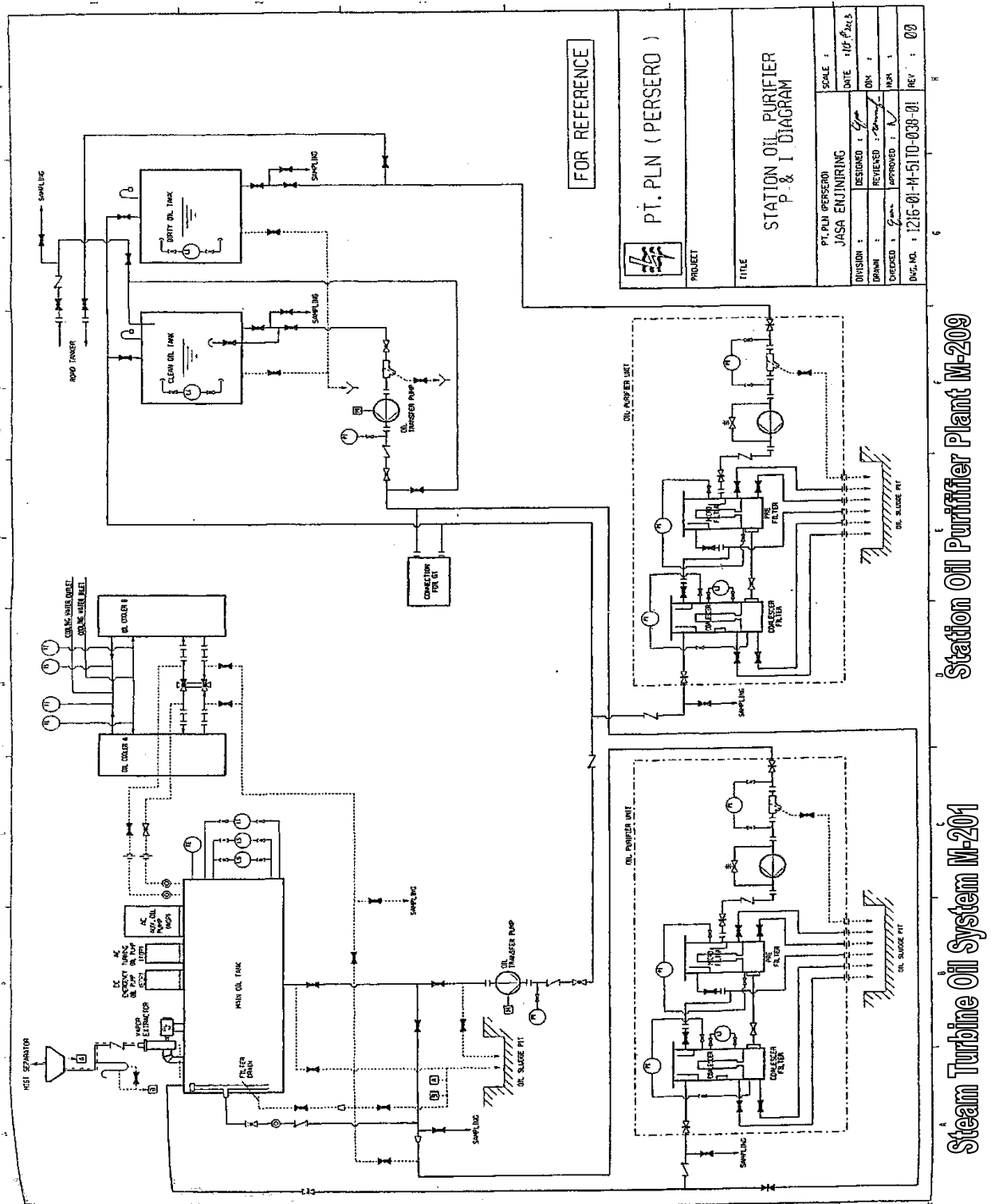
The Bidder shall furnish and install all electrical equipment and accessories required for the turbine lubricating oil purifier.

For general electrical requirements, see **Part 4 Section 3, E-001**.

8.2 Motors

For detailed electrical requirements, for Low Voltage Motors see **Part 4 Section 3, E-210** for Low Voltage Motors.

Figure 1 STATION OIL PURIFIER P&I DIAGRAM



(Last of page M-209)